

SHOULD-COST ANALYSIS REPORT

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Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

Seat Locking Bracket

Commodity: SHEET METAL · Currency: CNY · FX Rate: 9.0498 to GBP · Operations: 1 · Region: CN

Total Should-Cost	Material	Process	Labour	Margin
¥16.25	25.5%	0.7%	0.1%	7.4%

Model Confidence: **Low** · 1 traced operations · 4 data points auditable

Engine Warnings (1)

rawMaterial.materialId: Material rate confidence: Medium

Geometry Provenance — MEASURED (OCCT B-rep kernel)

Volume, weight and every feature were measured from the CAD solid by the Open CASCADE kernel. Measured volume 71.1 cm³; mass for the selected material family 0.558 kg.

Material cost and geometry-derived machining are grounded in the actual solid — not an estimate.

Key Assumptions

Annual volume: 200,000 · Region: CN · Commodity: sheet metal

Alloy / material: DC01 · Net weight: 0.558 kg (measured)

Material utilisation: 90% · Overhead: 12% of factory base · Margin: 8% of subtotal · Operations: 1

§1 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
1. Raw Material	¥4.14	25.5%	
2. Process (Machine)	¥0.11	0.7%	
3. Direct Labour	¥0.02	0.1%	
4. Tooling (amortised)	¥7.88	48.5%	
5. Packaging	¥0.63	3.9%	

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
6. Logistics	¥0.81	5.0%	
Factory Cost	¥13.59	83.6%	
7. Overhead (SG&A)	¥1.46	9.0%	
Subtotal	¥15.05	92.6%	
8. Supplier Margin	¥1.20	7.4%	
TOTAL SHOULD-COST	¥16.25	100.0%	

§2 — Commercial Parameters			
Overhead Rate	12.0%	Supplier Margin	8.0%
Packaging / part	¥0.63	Logistics / part	¥0.81
Total Tooling Cost	¥1576325.79	Amortisation Volume	200,000 parts

Geometric DFM / DFA - measured from the CAD model

1 issue(s) across 2 instance(s), from 110 measured feature(s) and 2 rule(s). Every issue names the B-rep faces that produced it and the published source of its threshold. No finding on this part has a modelled cost path; each says why.

[ADVISORY] Hole depth inconsistent with a sheet part (2 instances)

Worst: # 3.8 mm hole is 12.4 mm deep (3.3:1) — deeper than a punched sheet feature.
measured IdRatio 3.091-3.253:1 threshold > 3:1
Faces: 326, 327, 397, 398
Fix: Confirm this is a formed collar / extruded hole rather than a punched one — the two cost very differently. If it is a plain hole, the part may not be sheet metal.
Source: Sheet-metal design guidelines (widely published; e.g. Xometry / Protolabs sheet-metal design guides, and Machinery's Handbook press-working sections) - Punched features in thin material
Industry design-guide figures rather than a formal standard, stated as such so the threshold can be argued with directly.

DFA - handling & insertion (Boothroyd method, geometric half)

Handling 1.13s + insertion 7.8s = 8.93s, 2.98x the 3s ideal part.

+6.3s Fasteners approach from more than one direction — the part must be re-oriented or the fixture indexed during assembly (10 distinct hole axes across 18 holes)

Theoretical minimum part count, and therefore the Boothroyd design-efficiency index, requires the full assembly — this analysis covers one part, so no efficiency ratio is given.

Nesting and tangling risk is not assessed: it depends on how parts are presented in bulk, which the geometry alone does not determine.

Rotational symmetry is approximated from the bounding box, not from an inertia-tensor shape comparison.

What was NOT checked

No material family was confirmed, so any check whose threshold depends on the alloy or resin did not run.

Bend-to-hole distance was not checked: the kernel measures bend COUNT and total bend length but does not yet emit per-bend lines, so the distance from each hole to its nearest bend cannot be computed.

Flange length against tooling minimum was not checked, for the same reason.

Bend relief at notches was not checked — it needs per-bend endpoints.

Bend radius against material thickness was not checked — it needs the per-bend inner radius.

Blank utilisation was not checked — it needs a flat pattern, which the kernel does not unfold.

§3 — Material Detail

Parameter	Value	Unit	Notes
Material ID	mat-dc01	ID	
Grade / Specification	DC01		UK CR mild sheet, delivered small-lot. Index-anchored: C R C € 7 8 1 / t ! ' £ 0 . 6 7 / k g m i l l + £ 0 . 1 9 / k g s t o c k h o u s e p r e m i u m . B a s i s : F a s t m a r k e t s E U H R C € 6 9 1 / t M a y - 2 6 + € 9 0 C R C , F X 0 . 8 5 5 . Y i e l d ~ 1 4 0 M P a . R e g i o n a l a d j . × 0 . 8 8 0 (m i l l S t e e l)
Region	China		
Net Finished Weight	0.5580 kg	kg	Weight in finished part
Gross Weight (stock)	0.6181 kg	kg	Net ÷ utilisation ratio
Scrap / Runner Weight	0.0601 kg	kg	Gross - Net
Material Utilisation	90.3%	%	Benchmark: stamping/blanking 55–75 % (nesting + skeleton scrap)
Material Price	¥6.85	CNY /kg	UK CR mild sheet, delivered small-lot. Index-anchored: C R C € 7 8 1 / t ! ' £ 0 . 6 7 / k g m i l l + £ 0 . 1 9 / k g s t o c k h o u s e p r e m i u m . B a s i s : F a s t m a r k e t s E U H R C € 6 9 1 / t M a y - 2 6 + € 9 0 C R C , F X 0 . 8 5 5 . Y i e l d ~ 1 4 0 M P a . R e g i o n a l a d j . × 0 . 8 8 0 (m i l l S t e e l)
Scrap Recovery Price	¥1.59	CNY /kg	
Gross Material Cost	¥4.23	CNY	Gross × price/kg
Scrap Credit	-¥0.10	CNY	Scrap × recovery price
NET RAW MATERIAL COST	¥4.14	CNY	Gross - scrap credit + consumables
Data Confidence	Medium		
Effective Date	2026-07		

§4 — Operations Detail

4A Machine Operations

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
Press (transfer)	100T Mechanical Press (general press shop)	¥72.91	0.07	1	85.0%	¥0.11
TOTAL						¥0.11

4B Labour Detail

Operation	Labour Grade	Rate / hr	Man ning	Lab. min	Eff %	Labour Cost	Op Total
Press (transfer)	Semi-skilled Operator	¥49.77	0.25	0.07	95.0 %	¥0.02	¥0.12
TOTAL						¥0.02	¥0.12

4C Machined Features — Geometry Audit

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Face	13125 mm ²	2	Face milling (facing)	3.68	¥15.60	Yes
Face	7909 mm ²	2	Face milling (facing)	2.38	¥10.07	Yes
Face	7170 mm ²	2	Face milling (facing)	2.19	¥9.29	Yes
Face	2951 mm ²	2	Face milling (facing)	1.14	¥4.82	Yes
Face	591 mm ²	2	Face milling (facing)	0.55	¥2.32	Yes
Pocket	13125 mm ² × 15	2	Pocket milling (rough + finish)	66.03	—	No
Pocket	7909 mm ² × 18	1	Pocket milling (rough + finish)	24.43	—	No
Pocket	7909 mm ² × 12	1	Pocket milling (rough + finish)	15.80	—	No
Pocket	7170 mm ² × 22	1	Pocket milling (rough + finish)	27.09	—	No
Pocket	7170 mm ² × 8	1	Pocket milling (rough + finish)	9.47	—	No
Pocket	2951 mm ² × 22	1	Pocket milling (rough + finish)	11.67	—	No
Pocket	2951 mm ² × 7	1	Pocket milling (rough + finish)	3.92	—	No
Pocket	591 mm ² × 16	1	Pocket milling (rough + finish)	2.00	—	No
TOTAL costed		10		9.94	¥42.12	Yes

9 detected feature(s) NOT costed

9 machinable feature instance(s) (160.4 min) were detected in the geometry but are NOT included in this cost — on a near-net part they may be as-cast. If they are machined, tick them in the Machined-Features panel and re-cost; this could add up to ¥679.89.

\$5 — Machine Rate Buildup

100T Mechanical Press (general press shop) [press-100t] · Confidence: Low · UK general press-shop benchmark (standard/amortised mechanical press), Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥89592.76	¥32.00	3,500 hr/yr available
Maintenance	¥44796.38	¥16.00	
Energy	¥13771.39	¥4.92	
Floor Space	¥24886.88	¥8.89	
Indirect Support	¥19909.50	¥7.11	
Finance Cost	¥11199.10	¥4.00	
TOTAL — 100T Mechanical Press (general press shop)	¥204156.01	¥72.91	Effective hours: 2800/yr

How to read the machine rate

Rate/hr = (annual depreciation + maintenance + energy + floor + indirect + finance) ÷ effective hours, where effective hours = available hours × utilisation.

Effective hours encode the shift pattern: a lower hours-base raises £/hr and a higher one lowers it. A challenge from a supplier will target this divisor and whether depreciation is machine-only or full-cell (machine + dosing furnace + trim + robots/extraction) — state your basis when defending the number.

\$6 — Rate Traceability

Field	Value	Unit	Source / Reference	Rate ID	Conf.
material.pricePerKg	0.7568	£/kg	UK CR mild sheet, delivered small-lot. Index-anchored: CRC €781/t! £0.67/kg stockholder premium. Basis: Fastmarkets EU HRC €691/t May-26 +€90 CRC, FX 0.855. Yield ~140 MPa. Regional adj. ×0.880 (millSteel)	mat-dc01	Medium
material.scrapRecoveryPricePerKg	0.1760	£/kg	UK CR mild sheet, delivered small-lot. Index-anchored: CRC €781/t! £0.67/kg mill + £0.19/kg stockholder premium. Basis: Fastmarkets EU HRC €691/t May-26 +€90 CRC, FX 0.855. Yield ~140 MPa. Regional adj. ×0.880 (millSteel)	mat-dc01	Medium
Press (transfer).machineRatePerHr	8.0569	£/hr	UK general press-shop benchmark (standard/amortised mechanical press), Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	press-100t	Low
Press (transfer).labourRatePerHr	5.5000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-semiskilled	Low

§7 — Should-Cost with Uncertainty

Monte-Carlo · Low confidence

Estimate (± band)	P10 · optimistic	P50 · median	P90 · conservative	Band	CV
¥16.25 ± 29.4%	¥11.86	¥15.57	¥21.45	wide	23.9%

Range reflects how well each cost driver is known (High / Medium / Low confidence). Tighten it by attaching CAD, a BOM, or logging an actual quote.

Why confidence is Low

Main band drivers: 2 rate(s) at Low confidence; tooling amortisation carries the widest per-bucket spread; some detected features not costed (see §4C).

What's excluded from this unit cost

One-time NRE — tooling / die, fixtures and CNC programming — is amortised separately, not in this unit cost.
 Import duty and international freight (regional table is Ex-Works).
 Formal embodied-carbon reporting (figures are indicative cradle-to-gate — replace with supplier EPDs).

§8 — Sensitivity Analysis

± 10% driver swing · ranked by ¥ impact

Cost Driver	Base	-10% cost	+10% cost	Range ¥
Amortisation Volume	200000parts	+6.5%	-5.3%	¥1.93
Tooling Cost	¥1576325.79	-5.9%	+5.9%	¥1.91
Material: DC01	¥6.85/kg	-3.2%	+3.2%	¥1.02
Material utilisation	0.9	+2.7%	-2.2%	¥0.79
Overhead %	12%	-1%	+1%	¥0.31
Supplier Margin %	8%	-0.7%	+0.7%	¥0.24
Press (transfer): Cycle Time	0hr	-0.1%	+0.1%	¥0.03
Press (transfer): Machine Rate	¥72.91/hr	-0.1%	+0.1%	¥0.03

§9 — Regional Cost Comparison

Ex-Works · per-region should-cost · vs China

Region	Material	Process	Labour	Tooling	Overhead	Ex-Works	Logistics	Total	vs China
United Kingdom (GBP)	¥4.70	¥0.19	¥0.06	¥14.33	¥1.94	¥21.23	¥0.56	¥23.90	+47%
Germany (EUR)	¥4.84	¥0.20	¥0.10	¥15.05	¥2.24	¥22.43	¥0.65	¥25.30	+56%
France (EUR)	¥4.80	¥0.18	¥0.07	¥13.18	¥1.93	¥20.16	¥0.65	¥22.91	+41%
Spain (EUR)	¥4.70	¥0.17	¥0.04	¥12.61	¥1.68	¥19.20	¥0.65	¥21.88	+35%
Poland (PLN)	¥4.56	¥0.14	¥0.03	¥10.32	¥1.29	¥16.33	¥0.67	¥18.77	+15%
Turkey (TRY)	¥4.23	¥0.12	¥0.01	¥8.60	¥1.02	¥13.98	¥0.73	¥16.25	+0%
China (CNY)	¥4.14	¥0.11	¥0.02	¥7.88	¥1.46	¥13.60	¥0.81	¥16.25	Base
India (INR)	¥4.23	¥0.10	¥0.01	¥7.45	¥0.86	¥12.65	¥0.84	¥14.83	-9%
Mexico (MXN)	¥4.47	¥0.12	¥0.02	¥8.60	¥1.04	¥14.24	¥0.76	¥16.50	+2%
United States (USD)	¥4.70	¥0.17	¥0.08	¥12.18	¥1.64	¥18.76	¥0.70	¥21.40	+32%

Per-region should-cost using regional labour, energy and rate benchmarks. Tooling held fixed. Ex-Works — excludes import duty + international freight. Indicative — confirm with RFQ.

§10 — Embodied Carbon

1.40 kgCO2e · 2.51 /kg · cradle-to-gate

Material	1.19 kgCO2e	Material factor	2.10 kg/kg (Steel)
Process	0.12 kgCO2e	Grid intensity	0.550 kg/kWh · 0.22 kWh
Logistics	0.09 kgCO2e	Total	1.40 kgCO2e

Indicative cradle-to-gate factors — replace with supplier EPDs for formal reporting.

[WARNING] Tooling amortisation is a significant cost driver

Finding	Tooling at 48.5% (£0.87/part) exceeds the benchmark of 10%. This is volume-sensitive.
Impact	High · up to 10% potential saving
Action 1	Doubling volume to 400,000 parts halves tooling cost to £0.44/part
Action 2	Evaluate family tooling (multiple parts per die) to spread NRE across higher volumes

[OPPORTUNITY] Packaging & logistics cost is above average

Finding	Packaging + logistics at 8.9% of total cost. For high-volume parts this should be under 2%.
Impact	Low · up to 2% potential saving
Action 1	Evaluate returnable/reusable packaging (racks, dunnage trays) vs expendable packaging
Action 2	Consolidate shipment frequency — larger drops reduce per-unit logistics cost

[INFO] Material cost is below benchmark range

Finding	Material at 25.5% is below the typical 30–52% range. Conversion cost is the dominant driver.
Impact	Low
Benchmark	Material %: yours 25.5% vs industry 30–52%
Action 1	Focus optimisation efforts on process and labour efficiency

§12 — Cost-Reduction Opportunities (ranked by saving)

6 opportunities · combined ¥1.93/part (~12%)

Ranked on money per part against this costing. The combined figure is the root-sum-square of the top three, capped at 40% — overlapping actions do not save twice, so the column is deliberately not summed.

#	Category	Action	Why it is on the list	Save/p art	%	Timeframe	Risk
1	Tooling	Volume Increase to Dilute Tooling NRE	Tooling at 48.5% of part cost is volume-sensitive. Doubling volume halves per-part tooling cost.	¥1.63	10.0 %	Quick Win	Low
2	Tooling	Tooling Ownership and End-of-Life Terms	£174,184 of tooling sits behind this part. Who owns it, who maintains it, and what happens at end of programme are worth points of piece price — and all of it is negotiable before the tool is cut.	¥0.65	4.0%	Quick Win	Low
3	Tooling	Tool-Life Extension Programme	Tooling at 48.5% of part cost means every extra thousand shots of die life is money. Surface treatment and disciplined	¥0.65	4.0%	Medium Term	Low

#	Category	Action	Why it is on the list	Save/p art	%	Timeframe	Risk
			maintenance are the cheapest capacity there is.				
4	Design & Geometry	Design Review for DFM Compliance	Formal DFM review with manufacturing engineering to identify part features that increase cost without functional benefit.	¥0.81	5.0%	Quick Win	Low
5	Commercial & Sourcing	Annual Volume Re-Commitment for Better Pricing	Provide 12-month rolling volume forecast to supplier to secure better unit pricing and tooling amortisation.	¥0.49	3.0%	Quick Win	Low
6	Packaging & Logistics	Pack Density / Cube Utilisation Review	Packaging plus logistics is 8.9% of part cost. Freight is bought by the cubic metre — parts that nest, stack or ship in denser dunnage cut it directly.	¥0.43	2.7%	Quick Win	Low

§13 — Opportunity by Category

4 categories with an actionable lever

Category	Opportunities	Best single action	Best save/part	Category total (indicative)
Tooling	3	Volume Increase to Dilute Tooling NRE	¥1.63	¥2.93
Design & Geometry	1	Design Review for DFM Compliance	¥0.81	¥0.81
Commercial & Sourcing	1	Annual Volume Re-Commitment for Better Pricing	¥0.49	¥0.49
Packaging & Logistics	1	Pack Density / Cube Utilisation Review	¥0.43	¥0.43

§15 — Implementation Roadmap

Quick Wins — Implement Immediately

- Volume Increase to Dilute Tooling NRE
- Tooling Ownership and End-of-Life Terms
- Pack Density / Cube Utilisation Review
- Design Review for DFM Compliance
- Annual Volume Re-Commitment for Better Pricing